

A 4-colour problem for dense triangle-free graphs

Stephan Brandt

*Technische Universität Ilmenau, Fak. für Mathematik & Naturwissenschaften, Postfach 100 565,
D-98684 Ilmenau, Germany*

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Abstract

Triangle-free graphs of order n with minimum degree exceeding $n/3$ satisfy strong structural properties in several respects. Nevertheless, it is not known whether those graphs can have arbitrarily large chromatic number. We conjecture that these graphs are 4-colourable and give an affirmative answer for regular maximal triangle-free graphs satisfying the degree bound. Moreover, we show that the vertex transitive members of this class are 3-colourable. The indicated problem has a fractional appeal and we present several related conjectures in fractional terms. © 2002 Elsevier Science B.V. All rights reserved.

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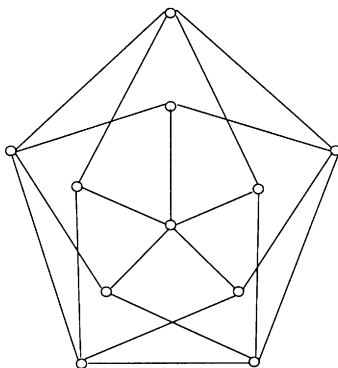
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1. Introduction and main results

Let $\mathcal{G}$ denote the class of triangle-free graphs with minimum degree exceeding $n/3$ where n denotes the number of vertices. These graphs have strong structural properties. The length of a longest cycle, for example, is as long as the independence number α admits, namely $\min\{n, 2(n - \alpha)\}$ [2], the independence number can be computed in polynomial time, based on the fact that for graphs in $\mathcal{G}$ that are maximal triangle-free (i.e. no edge can be added without creating a triangle) the independence number equals the maximum degree [3]. Moreover, Chen et al. [6] characterized the 3-colourable members of $\mathcal{G}$ in terms of homomorphisms and proved that the graphs in $\mathcal{G}$ with chromatic number at least 4 contain the Grötzsch graph (see Fig. 1) as an induced subgraph (for a simpler approach to a slightly stronger result, see [4]).

Conversely, Brandt and Pisanski [5] proved that there is a considerable wealth of 4-chromatic graphs in $\mathcal{G}$. Although so many properties of graphs in $\mathcal{G}$ are known, the

E-mail address: sbrandt@mathematik.tu-ilmenau.de (S. Brandt).

Fig. 1. The Grötzsch graph H_{11} .

question for the maximal chromatic number remains unsolved. The study was initiated by Erdős and Simonovits [7], who conjectured that the graphs in $\mathcal{G}$ are 3-colourable, and who presented a construction due to Hajnal showing that the chromatic number can be arbitrarily large within the class of triangle-free graphs with minimum degree exceeding $(1 - \varepsilon)n/3$ for every $\varepsilon > 0$. The conjecture was disproved by Häggkvist [9], presenting a 10-regular, 4-chromatic graph of order 29 based on the Grötzsch graph (see Fig. 1). Subsequently, Jin [11] conjectured that there are graphs in $\mathcal{G}$ with arbitrarily large chromatic number (for a more thorough treatment of the history, see [4]). In fact, we believe that the original conjecture of Erdős and Simonovits is true with 4 in place of 3.

Conjecture 1.1. Every graph in $\mathcal{G}$ is 4-colourable.

Our main result shows that this is the case for graphs in $\mathcal{G}$ that are additionally regular and maximal triangle-free.

Theorem 1.2. *Let G be a regular maximal triangle-free graph of order n with degree exceeding $n/3$. Then G is 4-colourable.*

In fact, we prove a stronger result which immediately implies Theorem 1.2.

Theorem 1.3. *Let G be a regular maximal triangle-free graph of order n with degree $d > n/3$. Then G contains a dominating star $K_{1,t}$ with $t \leq 3$.*

Obviously, the neighbourhoods of the vertices of the star form independent sets which cover $V(G)$. So $V(G)$ is covered by at most 4 independent sets, or, in other words, G is 4-colourable. If we additionally require that G is vertex transitive then we can show that G is 3-colourable.

Theorem 1.4. *Let G be a vertex transitive maximal triangle-free graph of order n with degree exceeding $n/3$. Then G is 3-colourable.*

In fact, we will be able to characterize these graphs (Corollary 4.4). All the maximal triangle-free graphs in $\mathcal{G}$ that we know are subgraphs of a regular maximal triangle-free supergraph in $\mathcal{G}$ that has the same chromatic number. The way how to obtain this supergraph is via the solution of a linear program, suggesting a fractional viewpoint towards the general problem. We will conclude this paper by presenting several further conjectures formulated in fractional terms that are related to the colouring problem and might indicate an approach towards a solution.

2. Concepts and notation

We will consider only finite, simple and undirected graphs (for the basic graph theoretical notation and concepts the reader is referred to introductory graph theory literature, e.g. [15]). A graph is triangle-free if it does not contain K_3 as a subgraph, and it is maximal triangle-free if no pair of non-adjacent vertices can be joined by an edge without creating a triangle. Observe that for a triangle-free graph, being maximal triangle-free is equivalent to having diameter at most 2. If G is a maximal triangle-free graph and v and w are vertices such that the neighbourhood $N(v) \subseteq N(w)$ then $N(v) = N(w)$. Indeed, otherwise we could add all missing edges joining v to $N(w)$ without creating a triangle. A pair of vertices v, w with $N(v) = N(w)$ will be called *similar* (in the literature, similar vertices are also called symmetric or twins). Similarity induces an equivalence relation on the vertex set $V(G)$. For a vertex v , let $S(v)$ denote its similarity class and $s(v) = |S(v)|$.

A *graph homomorphism* $G \rightarrow H$ is a mapping $\sigma: V(G) \rightarrow V(H)$, where $vw \in E(G)$ implies $\sigma(v)\sigma(w) \in E(H)$. Since a proper colouring of a graph is a partition into independent sets, a graph G is r -colourable if and only if $G \rightarrow K_r$. So graph homomorphisms can be seen as a refinement of graph colouring. Note that there is an injective homomorphism σ , if and only if G is a subgraph of H . Hence a bijective homomorphism is an isomorphism. An isomorphism from a graph to itself is an *automorphism* and a graph G is called *vertex transitive*, if for every pair (v, w) , $v, w \in V(G)$, there is an automorphism σ with $\sigma(v) = w$. A possibly surprising fact is that every graph G has a unique (up to isomorphism) minimal subgraph H to which it is homomorphic. This graph is called the *core* of G (see e.g. [10]). By the transitivity of the homomorphism relation, the core of a graph is its own core. We will call a graph that is its own core, a *core*. If G is maximal triangle-free and H is triangle-free then the only vertices that can be identified by a homomorphism from G to H are similar vertices.

The inverse concept to the identification of vertices by a homomorphism is vertex multiplication. The duplication of a vertex v is the addition of a new vertex v' joined to every neighbour of v and to no other vertex of G . Note that the final result of a sequence of vertex duplications only depends on the number of times a vertex is duplicated, but not on the order of vertex duplications. So we can express the resulting graphs by assigning the multiplicity $m(v)$ of v in the resulting graph, meaning the number of times v is multiplied (the case $m(v) = 0$ can be interpreted as the deletion

of v). If $m(v) = s$ for every $v \in V(G)$ then we denote the result by $G[\overline{K}_s]$ (this is a special case $H = \overline{K}_s$ of the lexicographic product $G[H]$).

A particularly interesting class of triangle-free graphs that has been rediscovered many times over the years are the graphs Γ_i probably first described by Andrásfai [1]. Let $V(\Gamma_i)$ be the corners of a regular $(3i - 1)$ -gon inscribed in a circle and let the edge set $E(\Gamma_i)$ consist of the pairs of vertices whose euclidean distance is bigger than the length of the side of a regular 3-gon inscribed in the circle. So $\Gamma_1 = K_2$, $\Gamma_2 = C_5$ and Γ_3 is the Möbius ladder with 4 spokes (sometimes also called Wagner graph).

The graphs Γ_i (and $\Gamma_i[\overline{K}_s]$ as well) belong to $\mathcal{G}$. These graphs are 3-colourable. Brandt and Pisanski [5] determined an infinite sequence of cores of 4-chromatic graphs in $\mathcal{G}$. They can be turned into graphs in $\mathcal{G}$ by suitable vertex multiplications. This sequence of graphs that was created interactively with a computer using the computer program Vega is called, due to the origin, Vega graphs.

For every $p \geq 11$, there exists a Vega graph $\mathcal{T}_p$ with p vertices which is obtained in the following way.

For $k \geq 4$ and $p = 3k + 1$ take a 6-cycle $C = (v_1, v_2, \dots, v_6)$, an edge $u_1 u_2$ and a copy of Γ_{k-2} with vertex set $\{w_1, w_2, \dots, w_{3k-7}\}$. Join u_1 to v_1, v_3, v_5 , u_2 to v_2, v_4, v_6 and w_{3i-j} to v_{3-j} and v_{6-j} for $1 \leq i \leq k - 2$ and $0 \leq j \leq 2$. The resulting graph is $\mathcal{T}_{3k+1}$. In order to obtain $\mathcal{T}_{3k}$ delete the vertex w_1 from $\mathcal{T}_{3k+1}$, and to obtain $\mathcal{T}_{3k-1}$ delete the vertex u_1 from $\mathcal{T}_{3k}$. Only the Grötzsch graph $\mathcal{T}_{11}$ and the graph $\mathcal{T}_{12}$ found by Jin [11] were previously known.

In the last section, we will use standard linear programming terminology that can be found in [12]. For more information about linear programming the reader is referred to Schrijver [13].

3. Basic observations

Let us first present some basic facts about triangle-free graphs in $\mathcal{G}$ that will be employed in the proofs. The first is a result due to Chen et al. [6] that separates the 3-colourable graphs in $\mathcal{G}$ from those with chromatic number at least 4.

Theorem 3.1 (Chen et al. [6]). *A graph $G \in \mathcal{G}$ is not homomorphic with a graph in Γ_i if and only if G contains the Grötzsch graph H_{11} as a subgraph.*

Note that the Grötzsch graph is 4-chromatic while the graphs in Γ_i are 3-colourable. An easy consequence is the following result, which was also observed in [6].

Corollary 3.2. *Let $i \geq 2$ be an integer and G be a 3-colourable triangle-free graph of order n with $\delta > n/(3i - 1)$. Then G is homomorphic with Γ_{i-1} .*

A basic fact that is frequently very helpful is the following statement.

Theorem 3.3 (Brandt [3]). *Let $G \in \mathcal{G}$ be maximal triangle-free. Then every maximum independent set of G is the neighbourhood of a vertex.*

In other words, the independence number $\alpha(G)$ equals the maximum degree $\Delta(G)$.

The central tool in the proof is the following forbidden subgraph statement, saying that a maximal triangle-free graph in $\mathcal{G}$ cannot contain the graph Q_3 of the 3-dimensional cube, which is $K_{4,4}$ minus a perfect matching, as an induced subgraph.

Lemma 3.4. *Let $G \in \mathcal{G}$ be maximal triangle-free. Then G does not contain Q_3 as an induced subgraph.*

With foresight to the fractional aspects in the last part, we will prove weighted generalizations of Theorem 3.3 and of Lemma 3.4.

For a graph G we call a weight function ϕ feasible if it assigns to every vertex of G a non-negative real value in such a way that the neighbourhood of every vertex receives total weight at least one. If we have a graph in $\mathcal{G}$ then the weight function ϕ assigning to each vertex weight $(\delta(G))^{-1}$ is a possible candidate of total weight less than 3.

Theorem 3.5. *Let G be a maximal triangle-free graph and let ϕ be a feasible weight function. If the total weight is less than 3 then every independent set of maximum weight of G is contained in the neighbourhood of a vertex.*

Let us first derive Theorem 3.3 from Theorem 3.5.

Proof of Theorem 3.3. Let $G \in \mathcal{G}$ be a maximal triangle-free graph and let ϕ be the weight function assigning to each vertex weight $(\delta(G))^{-1}$. Then G and ϕ satisfy the hypothesis of Theorem 3.5 and hence every maximum weight independent set belongs to the neighbourhood of a vertex. Since ϕ is constant this implies that every maximum independent set is contained in the neighbourhood of a vertex. $\square$

For proving Theorem 3.5 we need the fractional generalization of Lemma 3.4.

Lemma 3.6. *Let G be maximal triangle-free graph and let ϕ be a feasible weight function. If the total weight is less than 3 then G does not contain the cube Q_3 as an induced subgraph.*

Proof. Assume that G contains an induced Q_3 . Denote the vertices on one side of the bipartition of Q_3 by u_i and the vertices on the other side by w_i such that $u_i w_i \notin E(Q_3)$ for $1 \leq i \leq 4$. Since G has diameter 2, for every pair of vertices u_i, w_i there is a common neighbour v_i , which cannot be adjacent to any further vertex u_j, w_j ($j \neq i$). Since $\phi(V(G)) < 3$ and $\phi(N(v)) \geq 1$ for every $v \in V(G)$, there must be a vertex x that is adjacent to more than 4 vertices in $\bigcup_{1 \leq i \leq 4} \{u_i, v_i, w_i\}$. The indices may be chosen such that $u_1, w_1, v_2, \dots, v_4$ are the neighbours of x .

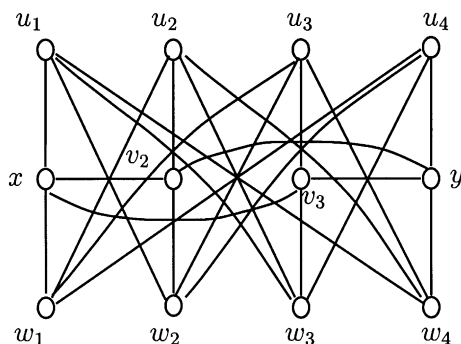


Fig. 2. The resulting subgraph of order 12.

Consider the subgraph H spanned by the 9 vertices $\bigcup_{2 \leq i \leq 4} \{u_i, v_i, w_i\}$. Since $\phi(V(G)) < 3$ and $\phi(N(v)) \geq 1$ for every $v \in V(G)$, there must be a vertex y of G being adjacent to 4 vertices of H which we may assume to be v_2, v_3, u_4, w_4 . Since G is triangle-free no vertex of G can be adjacent to more than four vertices in the subgraph H_{12} induced by $\{x, v_2, v_3, y\} \cup \bigcup_{1 \leq i \leq 4} \{u_i, w_i\}$ (see Fig. 2) and hence $\phi(V(G)) \geq \frac{1}{4} \sum_{v \in H_{12}} \phi(N(v)) \geq 3$, a contradiction. $\square$

Lemma 3.4 can be derived from Lemma 3.6 in the same way as Theorem 3.3 was derived from Theorem 3.5. Let us now prove Theorem 3.5.

Proof of Theorem 3.5. Suppose that S is an independent set of maximum weight that is not contained in the neighbourhood of a vertex. Let S' be a minimal subset of S that is not contained in the neighbourhood of a vertex. Since G has diameter 2 we get $s := |S'| \geq 3$ and, by the minimality of S' , for every vertex u_i of S' ($1 \leq i \leq s$) there is a vertex $w_i \in V(G)$ that is adjacent to all vertices of S' except u_i . If $s \geq 4$ then $\bigcup_{1 \leq i \leq 4} \{u_i, w_i\}$ induce a subgraph Q_3 and hence by Lemma 3.4 we have $\phi(V(G)) \geq 3$, a contradiction.

So we may assume $s = 3$. Let T_0 be the set of vertices of G having no neighbour in S' and T_2 be the vertices having exactly two neighbours. Since $\phi(V(G)) < 3$ and $\phi(N(v)) \geq 1$ for every $v \in V(G)$ and no vertex has 3 neighbours in S' , we have $\phi(T_2) > \phi(T_0)$. Since any two vertices of T_2 have a common neighbour in S' , the set T_2 is an independent set of larger weight than $S \subseteq T_0$, the final contradiction. $\square$

Another basic tool in the proofs of Theorems 1.2 and 1.4 is a slight strengthening of a result of Goddard and Kleitman [8], that they proved between the lines of their simple proof of Shi's Theorem [14], stating that the binding number of every triangle-free graph is less than $3/2$.

Lemma 3.7. *Let G be a maximal triangle-free graph in $\mathcal{G}$. If a, b are non-adjacent, non-similar vertices for which $|N(a) \cap N(b)|$ is as large as possible, then G contains*

an edge xy such that $xa, yb \in E(G)$ and all vertices in $N(a) \setminus N(b)$ are similar with x and all vertices in $N(b) \setminus N(a)$ are similar with y .

Proof. Since a and b are not similar, there must be a vertex $x \in N(a) \setminus N(b)$. Since G is maximal triangle-free, x must have a neighbour $y \in N(b) \setminus N(a)$. Assume that x has another neighbour $y' \in N(b) \setminus N(a)$ that is not similar with y . Then

$$\begin{aligned} d(b) + d(x) + d(y) + d(y') &\leq n + |N(x) \cap N(b)| + |N(y) \cap N(y')| \\ &\leq n + |N(x) \cap N(b)| + |N(a) \cap N(b)| \leq n + d(b) \end{aligned}$$

implying $\delta(G) \leq n/3$. Hence all vertices in $N(x) \cap N(b)$ are similar with y and, by a symmetric argument, also the vertices in $N(y) \cap N(a)$ are similar with x . Now assume that there is a vertex $x' \in N(a) \setminus N(b)$ that is not similar with x . By the similarity of the vertices in $N(y) \cap N(a)$, x' is not adjacent to y . By the similarity of the vertices in $N(x) \cap N(b)$, x, x' and b cannot have a common neighbour. Now x' must have a neighbour $y' \neq y \in N(b) \setminus N(a)$. So y' is not similar with y , yielding $xy' \notin E(G)$, and y, y' and a cannot have a common neighbour by the similarity of the vertices in $N(y) \cap N(a)$. So $\{a, x, y, b, y', x'\}$ induce a 6-cycle C and no vertex is adjacent to three vertices of C . We obtain

$$6\delta(G) \leq \sum_{v \in V(C)} d(v) \leq 2n,$$

implying $\delta(G) \leq n/3$. $\square$

Goddard and Kleitman [8] proved the weaker form replacing $N(a) \setminus N(b)$ by $N(a) \cap N(y)$ and $N(b) \setminus N(a)$ by $N(b) \cap N(x)$. They used this to prove that every maximal triangle-free graph has a vertex with $s(v) \geq 2d(v) + \delta(G) - n$ similar vertices. In fact, we believe that a global generalization of this statement is true. Recall that in a maximal triangle-free graph the vertices of the core correspond to the equivalence classes of similar vertices.

Conjecture 3.8. Let G be a maximal triangle-free graph of order n with minimum degree $\delta(G) > in/(3i - 1)$. Then its core G' has order at most $3i - 4$.

If true, the result would be best possible in view of the graphs Γ_i . For $2 \leq i \leq 10$ this can be derived from Theorem 3.1 and Corollary 3.2. Note that the truth would imply the slightly weaker form of Goddard and Kleitman's result, saying that every maximal triangle-free graph with $\delta > n/3$ has a set of $3\delta - n$ similar vertices, just because the average cardinality of a similarity class is at least that large.

If we have an equivalence class S of similar vertices then $\bigcap_{v \in N(S)} N(v) = S$. So we may ask for a minimal subset $T \subseteq N(S)$ such that $\bigcap_{v \in T} N(v) = S$. It turns out that in a maximal triangle-free graph in $\mathcal{G}$ we have $|T| \leq 3$. The proof technique resembles that of Theorem 3.5.

Lemma 3.9. *Let $G \in \mathcal{G}$ be maximal triangle-free and let S be a class of similar vertices of G . Then there is a set $T \subseteq N(S)$ with $|T| \leq 3$ such that $\bigcap_{v \in T} N(v) = S$.*

Proof. Choose a minimal subset $T \subseteq N(S)$ with $\bigcap_{v \in T} N(v) = S$. By the minimality of T for every vertex $v_i \in T$ there is a vertex $w_i \notin S$, being adjacent to every vertex of T , except v_i . If $|T| \geq 4$ then $\bigcup_{1 \leq i \leq 4} \{v_i, w_i\}$ induces a 3-dimensional cube Q_3 . By Lemma 3.4 we get that $\delta(G) \leq n/3$, thus $|T| \leq 3$. $\square$

4. Colouring regular graphs

Now we are prepared for proving the main results of Theorems 1.3 and 1.4.

Proof of Theorem 1.3. Choose a vertex v for which $s(v)$ is minimal and let $S = S(v)$. By Lemma 3.9 there must be a minimal subset $T = \{w_1, \dots, w_t\} \subseteq N(v)$ of cardinality $t \leq 3$ such that $\bigcap_{w \in T} N(w) = S$. We claim that $T' = \{v\} \cup T$ is a dominating set of G . Assume, to the contrary, that there is a set R of vertices which are not dominated by T' . If $t=1$ then all neighbours of v are similar and since G is maximal triangle-free vw_1 is a dominating edge. If $t=2$ then we have $n < d(v) + d(w_1) + d(w_2) \leq n - |R| + |S|$ implying $|R| < |S|$ and hence $R = \emptyset$ by the minimality of S . Finally assume that $t=3$. By the minimality of T , every pair of vertices w_i, w_j , $i \neq j$, has a common neighbour u_k in $G - S$, where $\{i, j, k\} = \{1, 2, 3\}$. By the definition of T , the vertices u_1, u_2, u_3 are distinct. Now $n + d < 4d = d(v) + d(w_1) + d(w_2) + d(w_3) \leq n - |R| + |U| + |S|$ where U is the set of vertices of G having at least two neighbours in T . Clearly, $U \supseteq S$ is an independent vertex set, so by Theorem 3.3 we have $|U| \leq d$. Rearranging terms we again get $|R| < |S|$, implying $R = \emptyset$. $\square$

So all maximal triangle-free graphs with minimum degree $\delta > n/3$ with chromatic number at least 5 are not regular, but, in view of Conjecture 1.1, we are possibly speaking about the empty set. On the other hand, in view of the Vega graphs, there is a moderately rich family of 4-chromatic maximal triangle-free graphs, which all have a regular supergraph with minimum degree $\delta > n/3$, obtained by vertex multiplications. We believe that actually a stronger statement than Conjecture 1.1 is true.

Conjecture 4.1. Every maximal triangle-free graph $G \in \mathcal{G}$ has a regular supergraph G' obtained from G by vertex multiplications.

The supergraph G' satisfies $\chi(G') = \chi(G)$ and in view of the fractional arguments below $\delta(G')/|G'| \geq \delta(G)/n$. So in view of Theorem 1.2, the validity of Conjecture 4.1 would imply the validity of Conjecture 1.1. Even more, there is still hope for a complete characterization of the maximal triangle-free graphs in $\mathcal{G}$ in terms of homomorphisms.

Problem 4.2. *Is there any maximal triangle-free graph with minimum degree $\delta > n/3$ whose core is neither a graph Γ_i nor a Vega graph T_i ?*

Even if there are further cores, eventually there might be a complete characterization of the possible cores of those graphs. The next result allows to characterize the vertex transitive members in the class of maximal triangle-free graphs in $\mathcal{G}$.

Theorem 4.3. *Let G be a d -regular maximal triangle-free graph in $\mathcal{G}$ with chromatic number at least 4. Then there are two vertices $y, z \in V(G)$ with $s(y) \neq s(z)$.*

Proof. First observe that any graph that does not contain two non-adjacent, non-similar vertices is complete multipartite. Since G has chromatic number at least 4, let a, b be two non-adjacent and non-similar vertices maximizing $|N(a) \cap N(b)|$. By Lemma 3.7 there is an edge xy with $xa, yb \in E(G)$, such that all the vertices in $N(a) \setminus N(b)$ are similar with x and all vertices in $N(b) \setminus N(a)$ are similar with y . If every vertex of G is adjacent to a, x or y , then G is 3-colourable. So there must be a vertex z that is adjacent to neither a, x, y nor b , since all vertices of $N(b) \setminus N(a)$ are in $N(x)$. Set $R = V(G) \setminus (N(a) \cup N(b) \cup N(x) \cup N(y))$. We have

$$\begin{aligned} n - |R| &\geq d(a) + d(b) + d(x) + d(y) \\ &\quad - |N(a) \cap N(b)| - |N(a) \cap N(y)| - |N(b) \cap N(x)| \\ &\geq d(b) + d(x) + d(y) - s(y) > n - s(y), \end{aligned}$$

implying $s(y) > |R| \geq s(z)$. $\square$

In particular, there cannot be an automorphism mapping y to z . We can even characterize the vertex transitive graphs in $\mathcal{G}$ based on the characterization of Chen, Jin and Koh (Theorem 3.1).

Corollary 4.4. *Every vertex transitive maximal triangle-free graph $G \in \mathcal{G}$ has the form $G \simeq \Gamma_i[\overline{K}_s]$ for integers $i, s \geq 1$.*

Proof. By Theorem 4.3 the graph G is 3-colourable since any automorphism maps vertices on vertices with the same cardinality of the similarity class. By Theorem 3.1 G is homomorphic with a graph Γ_i . Choose i to be minimal with this property. Note that the homomorphism is surjective, since $\Gamma_i - v$ is homomorphic with Γ_{i-1} for $i \geq 2$. Since G is maximal triangle-free this implies that G is obtained from Γ_i by vertex multiplications. Since the similarity class of each vertex needs to have the same cardinality, every vertex of G must be multiplied the same number of times, i.e. $G \simeq \Gamma_i[\overline{K}_s]$. $\square$

Since the graphs Γ_i , and therefore $\Gamma_i[\overline{K}_s]$ as well, are 3-colourable, Corollary 4.4 immediately implies Theorem 1.4.

5. Fractional aspects

Many graph invariants have a fractional variant that is frequently easier to handle, the linear programming relaxation of the original invariant. For each such invariant there is a dual invariant corresponding to the dual linear program. For a thorough introduction into the young field of fractional graph theory, the reader is referred to the recent monograph of Scheinerman and Ullman [12].

We investigate fractional invariants with respect to two different matrices associated with a graph G : the adjacency matrix A and the vertex-independent set incidence matrix B whose rows correspond to the vertices and whose columns correspond to the independent sets of G .

A *fractional colouring* of a graph G is an assignment of non-negative real weights to the independent sets such that the sum over the weights of the independent sets containing v is at least 1 for every vertex $v \in V(G)$. The fractional chromatic number $\chi_f(G)$ is the minimal weight sum of a fractional colouring, or, in other words, the value

$$\chi_f(G) = \min\{\mathbf{1}^T x \mid x \geq \mathbf{0}, Bx \geq \mathbf{1}\}. \quad (1)$$

Note that the chromatic number is the optimal solution of the corresponding integer program, where the entries of x are integers.

A *fractional clique* of a graph G is an assignment of non-negative real weights to the vertices such that the sum over the weights of the vertices in S is at most 1 for every independent set S of G . The fractional clique number $\omega_f(G)$ is the maximal weight sum of a fractional clique of G , or, in other words, the value

$$\omega_f(G) = \max\{\mathbf{1}^T y \mid y \geq \mathbf{0}, B^T y \leq \mathbf{1}\}, \quad (2)$$

The clique number is the optimal solution of the corresponding integer program. By the duality theorem of linear programming we have $\omega_f(G) = \chi_f(G)$ and by the previous reasoning

$$\omega(G) \leq \omega_f(G) = \chi_f(G) \leq \chi(G)$$

holds for every graph G . Note that we could as well restrict the columns of B to the maximal independent sets of G . In practise this is certainly sensible, but still, there may be exponentially many maximal independent sets, think for example of the graph consisting of k independent edges which has 2^k maximal independent sets. So it is not surprising that the computation of the fractional chromatic number is, in general, NP-hard.

A *total dominating set* is a subset $S \subseteq V(G)$ such that every vertex of $V(G)$ has a neighbour in S (this is also required for the vertices in S). The total domination number is the minimal cardinality of a total dominating set. If we replace the matrix B in the previous problem by the adjacency matrix A then the fractional chromatic number turns into the fractional total domination number

$$d_f(G) = \min\{\mathbf{1}^T x \mid x \geq \mathbf{0}, Ax \geq \mathbf{1}\}. \quad (3)$$

The dual linear program

$$D_f(G) = \max\{\mathbf{1}^T y \mid y \geq \mathbf{0}, Ay \leq \mathbf{1}\}, \quad (4)$$

(note that A is symmetric) having the same value, is usually referred to as the fractional open neighbourhood packing number (see [12]). We look at the parameter $d_f(G)$ from a different point of view: as the fractional variant of the inverse of the degree ratio. Note that for every graph G of order n

$$n/\delta(G) \geq d_f(G) = D_f(G) \geq n/\Delta(G).$$

Since both (3) and (4) have rational optimal solutions x and y , by multiplying x (y , resp.) with the least common multiple m of the denominators, we get integer solutions of the equation $Ax \geq \mathbf{m}$ ($Ay \leq \mathbf{m}$, resp.), where $\mathbf{m}$ is the all- m -vector. Multiplying the vertex v_i x_i -times (y_i times, resp.), we get a graph which has minimal (maximal, resp.) degree ratio $1/d_f(G)$. Another interpretation of $1/d_f(G)$ which is particularly interesting for us is that it is the supremum of the minimum degree ratio taken over all graphs H obtained from G by vertex multiplications. In this way, we can bound the minimum degree ratio of a whole class of graphs from above. It should be mentioned that $1/d_f(G)$ is at the same time the infimum of the maximum degree ratio $\Delta(H)/|H|$ over all graphs H obtained from G by vertex multiplications.

What is the link between fractional colouring and fractional total domination? If we have a triangle-free graph G then the neighbourhood of every vertex is an independent set. This means that the adjacency matrix A is a submatrix of B implying $d_f(G) \geq \chi_f(G)$. Anyway, there is an essential difference between $d_f(G)$ and $\chi_f(G)$. While $\chi_f(H) \leq \chi_f(G)$ for every subgraph H of G , there is no general relation between $d_f(H)$ and $d_f(G)$. The graphs in $\mathcal{G}$ that we are interested in satisfy $\chi_f(G) \leq d_f(G) \leq n/\delta(G) < 3$, so they are fractionally (< 3)-colourable.

Some of the previous problems and conjectures can be reformulated in terms of fractional parameters. At first sight, some of them do not seem to be related, but all of them would be settled in the affirmative if Problem 4.2 had a negative answer. Let us start with a relaxation of Conjecture 4.1 in terms of a linear programming problem, which turns out to be equivalent.

Conjecture 5.1. If G is a maximal triangle-free graph with $d_f(G) < 3$ then $A(G)x = \mathbf{1}$ has a rational solution $x \geq \mathbf{0}$.

Note that the solution x is a solution of both linear programs (3) and (4) and therefore optimal for both. In fact, for G being a core we get $x^* > \mathbf{0}$ (componentwise) for every optimal solution x^* . Assume otherwise and let G be a smallest order counterexample, satisfying $d_f(G) < 3$ and $A(G)x = \mathbf{1}$ for a rational vector $x \geq \mathbf{0}$ but $x_i = 0$ for an index i . If there are two vertices u and w in the neighbourhood of the corresponding vertex v_i , having no common neighbour except v_i , then $N(u)$, $N(w)$ and $N(v_i)$ are

disjoint except for v_i and

$$d_f(G) \geq \sum_{v_j \in N(v_i) \cup N(u) \cup N(w)} x_j = 3,$$

a contradiction. So every pair of neighbours of v_i has a further common neighbour and hence $G' = G - v_i$ is maximal triangle-free. On the other hand, G' and the vector x' , being x with entry x_i deleted, satisfy $A(G')x' = \mathbf{1}$, $x' > 0$ and $d_f(G') = d_f(G) < 3$. Considered as a weight function on the vertices, the vector x' and G' fulfil the hypothesis of Theorem 3.5. Therefore every maximum weight independent set is contained in the neighbourhood of a vertex but $N(v_i)$ is an independent set of maximum weight. So there must be a vertex v_j of G' with $N(v_j) \supseteq N(v_i)$ and since G' is maximal triangle-free, $N(v_j) = N(v_i)$, a contradiction to G being a core.

Since we did not succeed to answer Conjecture 5.1, it seems reasonable to formulate two further conjectures, which are possibly easier than Conjecture 5.1 since they immediately follow from Conjecture 5.1 while Conjecture 5.1 can be derived from a combination of both.

Recall that for a triangle-free graph G the fractional chromatic number $\chi_f(G) \leq d_f(G)$. For the graphs that we are interested in we believe that both fractional parameters are equal.

Conjecture 5.2. For every maximal triangle-free graph G with $d_f(G) < 3$ we have $d_f(G) = \chi_f(G)$.

Note that the vector $x \geq \mathbf{0}$ from Conjecture 5.1 is a solution of both (3) for $d_f(G)$ and of (2) for $\omega_f(G)$, again employing Theorem 3.5 as in the reasoning after Conjecture 5.1. Since $d_f(G) \geq \chi_f(G) = \omega_f(G)$ the vector x is an optimal solution for both and hence $d_f(G) = \omega_f(G) = \chi_f(G)$.

Let us now turn to the next conjecture, which states that the cores of our graphs are critical with respect to the fractional chromatic number.

Conjecture 5.3. If G is a maximal triangle-free core with $d_f(G) < 3$ then $\chi_f(G') < \chi_f(G)$ for every proper induced subgraph $G' < G$.

The truth of Conjecture 5.3 easily follows from Conjecture 5.1. Let G be a maximal triangle-free core with $d_f(G) < 3$ and write the matrix $B' = B(G')$ as an upper left submatrix of $B = B(G)$. Then any solution y' of $B'^T y' = \mathbf{1}$ gives rise to a solution y'' of $B^T y \leq \mathbf{1}$, where y'' is obtained from y' by appending 0's. On the other hand, the solution $x^* > \mathbf{0}$ of Conjecture 5.1 for cores gives rise to an optimal solution x^{**} of (1) with $Bx^{**} = \mathbf{1}$ by appending 0's. Assume that y'' is an optimal solution. Then $A(G)y'' = \mathbf{1}$ by complementary slackness, a contradiction since $y > \mathbf{0}$ does not hold (cf. the reasoning for cores after Conjecture 5.1).

Now assume that conversely we have a maximal triangle-free graph G satisfying $d_f(G) < 3$. Let G' be the core of G . Recall that the core is obtained from G by deleting

similar vertices. Assume that both Conjectures 5.2 and 5.3 are true. Then by Conjecture 5.2 every optimal solution of (2) for $\omega_f(G')$ satisfies $y > \mathbf{0}$, otherwise the graph $G' - v_i$ satisfies $\omega_f(G' - v_i) = \omega_f(G')$, where v_i is the vertex corresponding to a coordinate $y_i = 0$, contradicting the criticality of G' from Conjecture 5.3. Now by complementary slackness, every optimal solution x of (1) for $\chi_f(G')$ satisfies $B(G')x = \mathbf{1}$. Since by Conjecture 5.2 $d_f(G') = \chi_f(G')$, an optimal solution x^* of (3) for $d_f(G')$ extended by 0's is an optimal solution of (1) for $\chi_f(G')$. For this vector x^{**} , we get $B(G')x^{**} = \mathbf{1}$ and hence $A(G')x^* = \mathbf{1}$. Adding 0's to x^* we obtain a vector $x \geq \mathbf{0}$ which satisfies $A(G)x = \mathbf{1}$, implying the validity of Conjecture 5.1.

Certainly, we would prefer to determine $d_f(G)$ precisely, and since linear programming is polynomial, $d_f(G)$ is computable in polynomial time as well (in contrast to the matrix B for fractional chromatic number, the matrix A has polynomial size). But in many cases we want to find a bound on $\delta(G)$ only by the existence of a certain subgraph. So our aim is to minimize $d_f(G)$ over all triangle-free supergraphs of a given triangle-free graph H . If $d_f(H) = \chi_f(H)$ then the minimum is simply $d_f(H)$, but even for triangle-free supergraphs of the Petersen graph H_{10} this problem seems to be open. We have an upper bound of $\frac{16}{5}$ coming from the Clebsch graph, while $\frac{29}{10}$ is a lower bound which can be derived from Theorem 3.1. Note that this parameter is strictly between $\chi_f(H)$ and $d_f(H)$. Let us formulate the problem in a more popular non-fractional form.

Problem 5.4. *Is there a triangle-free graph of order n with minimum degree $\delta > 5n/16$ containing the Petersen graph?*

We can determine this parameter precisely for both the Petersen graph with one edge contracted H_{10}/e and the Petersen graph with one vertex deleted $H_{10} - v$. In both cases the parameter is $\frac{29}{10}$ while d_f equals 3 in both cases and $\chi_f(H_{10}/e) = \frac{11}{4}$ and $\chi_f(H_{10} - v) = \frac{5}{2}$, respectively.

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